

POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Near-Minimax Weakly Adaptive Regret Bounds in Adversarial Linear Bandits

Tesi di Laurea Triennale in Ingegneria Matematica

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Academic Year: 2025–2026

Abstract

Sequential decision making studies how a learner can repeatedly choose actions under uncertainty while adapting to the information revealed over time, with the *regret* providing a standard criterion for measuring performance against a suitable benchmark. We study *weakly adaptive* regret minimization for adversarial linear bandits on general compact convex domains, where the learner observes only the scalar loss of the chosen action. A regret guarantee over the entire horizon can conceal poor performance on shorter periods, motivating comparison with the best fixed action on every interval. Under an oblivious adversary, we analyze a continuous extension of exponential weights with *Löwner-John* exploration and uniform fixed-share. For dimension d, horizon T, and losses bounded in absolute value by L, we establish an expected interval regret bound of $O(dL\sqrt{T \log T})$. The analysis identifies the comparator's estimation error as the obstacle to a direct high-probability extension. By incorporating a pointwise variance correction into the exponential update, we obtain a regret bound of $O\Big(L\sqrt{dT\left(d\log T + \log \frac{T}{\delta}\right)}\Big)$ simultaneously over all intervals with probability at least $1-\delta$, for any $\delta \in (0,1)$. Both guarantees are sublinear in the total horizon and use parameters independent of the interval, so the learner requires no advance knowledge of the periods on which its performance will be evaluated. The results assume exact sampling and exact computation of second-moment matrices.

Keywords: Online Linear Optimization, Adversarial Bandit Feedback, Weakly Adaptive Regret Minimizers, High-Probability Interval Regret Bounds, Exponential Weights Algorithms.

Sommario

Il *sequential decision making* studia come un agente possa compiere ripetutamente delle azioni in condizioni di incertezza, adattandosi alle informazioni osservate nel tempo, con il *regret* che costituisce un criterio standard per valutarne le prestazioni rispetto a un opportuno benchmark. Studiamo la minimizzazione del regret *weakly adaptive* nel problema degli adversarial linear bandits su domini convessi compatti generali, nel quale l'algoritmo osserva soltanto la perdita scalare associata all'azione scelta. Una garanzia di regret sull'intero orizzonte temporale può nascondere prestazioni insoddisfacenti su periodi più brevi, motivando il confronto con la migliore azione fissa su ciascun intervallo. Assumendo che l'avversario scelga la sequenza delle perdite prima dell'interazione, analizziamo un'estensione continua di exponential weights con esplorazione di *Löwner-John* e uniform fixed-share. Indicando con d la dimensione, con T l'orizzonte temporale e con L un limite superiore al valore assoluto delle perdite, dimostriamo un bound sul regret intervallare atteso pari a $O(dL\sqrt{T \log T})$. L'analisi individua nell'errore di stima della perdita del comparatore l'ostacolo a un'estensione diretta in alta probabilità. Introducendo nell'aggiornamento esponenziale una correzione puntuale basata sulla varianza, otteniamo un bound di regret pari a $O\Big(L\sqrt{dT\left(d\log T + \log\frac{T}{\delta}\right)}\Big)$, valido simultaneamente su tutti gli intervalli con probabilità almeno $1-\delta$, per ogni $\delta \in (0,1)$. Entrambe le garanzie sono sublineari nell'orizzonte totale e utilizzano parametri indipendenti dall'intervallo, senza richiedere la conoscenza anticipata dei periodi sui quali verranno valutate le prestazioni. I risultati assumono campionamento esatto e calcolo esatto delle matrici dei secondi momenti.

Parole chiave: Ottimizzazione Lineare Online, Feedback Bandit Avversariale, Bound del Regret Intervallare in Alta Probabilità, Algoritmi a Pesi Esponenziali.

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3.1 Löwner position for a cubic decision set in $\mathbb{R}^3$. The transformed set $\mathcal{L}(\mathcal{K})$ is inscribed in the Euclidean unit ball $\mathbb{B}_2^3$, which represents its minimum volume enclosing ellipsoid. The vertices $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_8$ are contact points with $\partial\mathbb{B}_2^3$ and support a probability distribution yielded by $\pi_1, \ldots, \pi_m$. The arrow in red illustrates an exploratory action toward a contact point, starting from $\boldsymbol{x}_0$ (the barycenter of $\mathbb{B}_2^3$ and the coordinate origin). This construction encourages exploration on geometrically informative boundary directions. . 10

3.2 Intuitively, uniform fixed-share helps the learner revisit downweighted regions through uniform draws, whose mean is the barycenter of $\mathcal{K}$. From the common initialization $\boldsymbol{x}_0$, the schematic trajectories $\boldsymbol{x}'_1, \ldots, \boldsymbol{x}'_5$ (bordeaux, with fixed-share) and $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_4$ (blue, without fixed-share) initially move towards the comparator $\boldsymbol{u}$ and then turn towards $\boldsymbol{u}'$ after a loss change. The fixed-share trajectory shows a conservative response to the initial losses, reflecting the goal of attaining lower interval regret and comparable static regret. 12

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1 | Introduction

In linear bandits, the learner selects an action from a compact convex set and observes only its scalar loss. We study the adversarial setting, in which the sequence of linear losses is chosen without distributional assumptions. The model encompasses settings such as network routing with aggregate costs [10] and portfolio allocation, where the return observed in each period is linear in the investment weights [9]. Classical guarantees compare cumulative loss with that of the best fixed action over the entire horizon [6, 10]. Such a comparison can hide poor performance on shorter periods: losses relative to a fixed action on one interval may be compensated by gains on another. This motivates guarantees on every interval, where the best fixed action may differ from one interval to the next. We study *weak adaptivity*, which requires a regret bound sublinear in the total horizon T on every interval, using the same algorithm throughout. This requirement makes no assumptions about when the actions that perform well may change. A possible extension to this work concerns primal-dual methods for constrained online learning [4, 8]. If the primal payoffs admit linear bandit feedback, an interval regret algorithm could provide an explicit primal subroutine, provided its guarantees extend to the history-dependent losses generated by the dual learner.

Our contribution. We analyze two continuous versions of exponential weights with John's exploration and uniform fixed-share, following Luo et al. [22]. For dimension d and loss bound L, Theorem 4.1 gives expected interval regret $O(dL\sqrt{T \log T})$ with parameters fixed across intervals. With the pointwise correction of Zimmert and Lattimore [26], Theorem 4.2 gives a bound of $O\Big(L\sqrt{dT\big(d \log T + \log \frac{T}{\delta}\big)}\Big)$ simultaneously on all intervals with probability at least $1 - \delta$. The proofs combine comparator smoothing, interval potentials, and martingale concentrations. We assume an oblivious adversary, exact sampling and second moments. Efficient implementation and extension to adaptive losses require more work, as discussed later. Chapters 2-4 present model, algorithms and proofs. Both guarantees attain the worst-case expected minimax scale in dimension and horizon up to logarithmic factors; we do not claim sharp dependence on the confidence parameter.

1.1. Related Work

Static regret. Auer et al. [2] establish expected and high-probability regret guarantees for multi-armed bandits through EXP3 and EXP3.P. For linear bandits, where exploration must also reflect the geometry of the action set, Dani et al. [10] develop exponential weights methods, while Bubeck et al. [6] combine EXP2 with John's exploration to obtain nearly minimax expected regret. High-probability bounds require further control of estimation errors: Bartlett et al. [3] use biased estimates, Lee et al. [21] employ barrier regularization, and Zimmert and Lattimore [26] develop several constructions, including corrected exponential weights. These guarantees compare the learner with a fixed action over the full horizon and do not directly control regret on intervals of arbitrary length. Shamir [24] establishes worst-case regret lower bounds of order $\Omega(d\sqrt{T})$ on convex domains under subgaussian losses, which extend to uniformly bounded losses up to logarithmic factors. Since the full horizon is itself an interval, the same lower bounds apply to weakly adaptive regret.

Interval regret. Hazan and Seshadhri [15] study regret guarantees on arbitrary intervals. Fixed-share, introduced to track changing comparators [16], retains weight on actions penalized in earlier rounds; Neu combines it with implicit exploration in EXP3-SIX [23]. Castiglioni et al. [8] establish a high-probability interval guarantee for this algorithm and use weakly adaptive minimizers under budget and return-on-investment constraints. Bernasconi et al. [4] extend the primal-dual approach to general long-term constraints. For linear bandits, Luo et al. [22] analyze switching regret and outline EXP2 with fixed-share on compact convex domains. Switching regret uses piecewise constant actions, whereas dynamic regret permits a comparator that changes at every round [17, 27]. Strongly adaptive bounds would instead depend on interval length, but polynomially sublinear rates of this form are impossible under general bandit feedback [11]. We therefore seek horizon-dependent bounds that hold on every time interval, regardless of its starting point.

Computational considerations. Discretization can make exponential weights costly on high-dimensional continuous domains. A continuous implementation must address two computational challenges: sampling and exploration design. Without the quadratic correction, fixed share produces mixtures of log-concave components, allowing theoretically efficient but costly approximate sampling. The correction used in our interval analysis generally destroys this structure, so the same sampling guarantees no longer apply directly. Constructing the John exploration design with the required variance control also depends on the representation of $\mathcal{K}$: explicit vertex descriptions permit polynomial-time approximation, whereas separation access alone does not ensure the same guarantee [6, 25]. Our

analysis assumes access to this design, exact sampling, and second moments; an efficient implementation of the corrected algorithm on general convex domains remains open.

Algorithm or reference	Problem	Expected regret	High-probability regret	Scope	Efficient
EXP3 / EXP3.P [2]	MAB	$\widetilde{O}(\sqrt{KT})$	$\widetilde{O}(\sqrt{KT})$	Static	Yes
EXP3-SIX [23]	MAB	–	$\widetilde{O}(\sqrt{KST})$	Switching	Yes
EXP3-SIX [8]	MAB	–	$\widetilde{O}(\sqrt{KT})$	Interval	Yes
SCRiBLe [1]	Linear	$\widetilde{O}(d^{\frac{3}{2}}\sqrt{T})$	–	Static	Yes
Bartlett et al. [3]	Linear	–	$\widetilde{O}(d^{\frac{3}{2}}\sqrt{T})$	Static	No*
EXP2 + John's exploration [6]	Linear	$\widetilde{O}(d\sqrt{T})$	–	Static	No*
Volumetric spanners [14]	Linear	$\widetilde{O}(d\sqrt{T})$	–	Static	Yes
Bias No More [21]	Linear	–	$\widetilde{O}(d^{\frac{7}{2}}\sqrt{T})$	Static	Yes
Return of the Bias (log-det) [26]	Linear	–	$\widetilde{O}(d\sqrt{T})$	Static	Unknown
Return of the Bias (entropic barrier) [26]	Linear	–	$\widetilde{O}(d^2\sqrt{T})$	Static	Yes
Return of the Bias (EXP2) [26]	Linear	–	$\widetilde{O}(d\sqrt{T})$	Static	No*
Corralling [22]	ℓ_p balls, $p \in (1, 2]$	$\widetilde{O}(\sqrt{dST})$	–	Switching	Yes
EXP2 + fixed-share [22]	Linear	$\widetilde{O}(d\sqrt{ST})^{\dagger}$	–	Switching	Yes‡
This work: Algorithm 3.1	Linear	$\widetilde{O}(d\sqrt{T})$	–	Interval	–
This work: Algorithm 3.2	Linear	–	$\widetilde{O}(d\sqrt{T})$	Interval	–

Table 1.1: Selected guarantees for adversarial bandits with unit loss scale. Here K counts arms, d is the dimension, T is the horizon, and S counts comparator segments. Linear denotes general linear bandit domains; Interval denotes weak adaptivity. Logarithmic and confidence factors are suppressed, and a dash indicates that no claim is made in that column. Efficiency is relative to the stated oracle assumptions. *Discretization may require exponentially many actions. †The switching bound is stated without proof. ‡Theoretically polynomial-time sampling may be costly. The rates for this work assume exact sampling and moments; Chapter 4 states the precise bounds and assumptions for both algorithms.

1.2. Notation

$c, \boldsymbol{v}, \boldsymbol{A}$	Scalar; vector in $\mathbb{R}^d$; matrix in $\mathbb{R}^{d\times d}$.
$\langle \boldsymbol{v}, \boldsymbol{w}\rangle$	Inner product $\boldsymbol{v}^\top \boldsymbol{w}$.
$\boldsymbol{v} \otimes \boldsymbol{w}$	Outer product $\boldsymbol{v}\boldsymbol{w}^\top$.
$\boldsymbol{I}_d$	Identity in $\mathbb{R}^d$.
$[T]$	Set of integers $\{1, \dots, T\}$.
$\mathcal{B}(\mathcal{S})$	Borel σ-algebra of $\mathcal{S} \subseteq \mathbb{R}^d$.

$\mathrm{conv}(\mathcal{S})$	Convex hull of $\mathcal{S}$.
$\mathrm{vol}(\mathcal{S})$	Volume of $\mathcal{S}$.
$\|\boldsymbol{x}\|_2$	Euclidean norm $\sqrt{\langle \boldsymbol{x}, \boldsymbol{x}\rangle}$.
$\mathbb{B}_2^d$	Euclidean ball centered the origin.
$\|\boldsymbol{x}\|_{\boldsymbol{A}}$	$\boldsymbol{A}$-induced norm $\sqrt{\langle \boldsymbol{x}, \boldsymbol{A}\boldsymbol{x}\rangle}$.
$\|\boldsymbol{x}\|_t$	For a sequence of geometries $(\boldsymbol{A}_1, \dots, \boldsymbol{A}_T)$
$\boldsymbol{A} \succeq \boldsymbol{B}$	The matrix $\boldsymbol{A} - \boldsymbol{B}$ is positive semidefinite.

1.3. Useful Facts

Elementary inequalities. For $a \le 1$, $b > -1$ and $c, d \in \mathbb{R}$, it holds

$$e^a \le 1 + a + a^2, \qquad \log(1+b) \le b, \qquad (c-d)^2 \le 2c^2 + 2d^2. \tag{1.1}$$

Quadratic forms and conditional trace. For $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^d$, $\langle \boldsymbol{v}, \boldsymbol{w}\rangle^2 = \langle \boldsymbol{v}, (\boldsymbol{w} \otimes \boldsymbol{w})\boldsymbol{v}\rangle$. This allows us to express conditional quadratic moments as traces. If $\boldsymbol{z}$ is square-integrable and $\boldsymbol{A}$ is an $\mathcal{F}$-measurable symmetric matrix for which the terms are integrable, we have

$$\mathbb{E}[\langle \boldsymbol{z}, \boldsymbol{A}\boldsymbol{z}\rangle \mid \mathcal{F}] = \mathrm{tr}(\boldsymbol{A}\,\mathbb{E}[\boldsymbol{z} \otimes \boldsymbol{z} \mid \mathcal{F}])\,. \tag{1.2}$$

In particular, if $\boldsymbol{M} = \mathbb{E}[\boldsymbol{z} \otimes \boldsymbol{z} \mid \mathcal{F}]$ is positive definite, then $\mathbb{E}\left[\|\boldsymbol{z}\|_{\boldsymbol{M}^{-1}}^2 \mid \mathcal{F}\right] = d$.

Conditional sampling. Let q_t denote the conditional law of $\boldsymbol{x}_t$. For a function f_t determined by $\mathcal{F}_{t-1}$, averaging over the current action is equivalent to conditional expectation

$$\mathbb{E}_{t-1} f_t(\boldsymbol{x}_t) := \mathbb{E}[f_t(\boldsymbol{x}_t) \mid \mathcal{F}_{t-1}] = \mathbb{E}_{\boldsymbol{x} \sim q_t} f_t(\boldsymbol{x}). \tag{1.3}$$

Estimator moments. The estimator in Chapter 3 is conditionally unbiased

$$\mathbb{E}_{t-1}\widehat{\boldsymbol{\ell}}_t = \boldsymbol{M}_t^{-1}\mathbb{E}_{t-1}[\boldsymbol{x}_t \otimes \boldsymbol{x}_t]\boldsymbol{\ell}_t = \boldsymbol{\ell}_t. \tag{1.4}$$

Moreover, for every $\boldsymbol{x} \in \mathcal{K}$, the bound $|y_t| \le L$ controls its pointwise second moment

$$\mathbb{E}_{t-1}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle^2 \le L^2 \langle \boldsymbol{x}, \boldsymbol{M}_t^{-1}\mathbb{E}_{t-1}[\boldsymbol{x}_t \otimes \boldsymbol{x}_t]\boldsymbol{M}_t^{-1}\boldsymbol{x}\rangle = L^2 \|\boldsymbol{x}\|_t^2. \tag{1.5}$$

2 | Problem Formulation

In this chapter, we formalize the adversarial linear bandit problem and state the assumptions on the decision set, losses, and available feedback. We define randomized strategies through probability kernels, making their dependence on past observations explicit. We then introduce regret benchmarks and distinguish guarantees in expectation from those holding with high probability. Particular emphasis is placed on regret over arbitrary time intervals, which forms the basis for the notion of weak adaptivity analyzed in this thesis.

2.1. Adversarial Linear Bandits

The online learning problem can be viewed as a *repeated zero-sum game* between a *learner* and an *environment*. At each round $t \in [T]$, the learner plays an *action* $\boldsymbol{x}_t$ and the environment selects an *evaluation function* λ_t. Depending on the adopted convention, λ_t represents either a *loss* to be minimized or a *utility* to be maximized; throughout this work, we adopt the loss-minimization convention. The learner then observes a feedback signal $y_t = y(\boldsymbol{x}_t, \lambda_t)$ determined by the assumed observation model. We consider *bandit feedback*, under which only the realized value $\lambda_t(\boldsymbol{x}_t)$ is observed, while the function λ_t itself remains unknown to the learner. At each round, the interaction is summarized as follows:

1. the learner plays $\boldsymbol{x}_t$ according to its strategy;
2. the environment selects λ_t and reveals $\lambda_t(\boldsymbol{x}_t)$;
3. the learner observes $y_t = \lambda_t(\boldsymbol{x}_t)$ and updates its strategy.

Depending on the structure of the learner's action set and the loss class, different bandit models arise. For instance, *multi-armed bandits* have finite or countable action sets, whereas *linear bandits* allow compact convex action sets while restricting the losses to linear functions. We study the latter in the *adversarial* setting, meaning that no distributional assumptions are imposed on the loss sequence, in contrast to stochastic bandits [5]. Formally, let $\mathcal{K} \subset \mathbb{R}^d$ be a non-empty, compact, full-dimensional convex set. We assume that the environment acts as an *oblivious adversary*: it fixes in advance the sequence of loss vectors $(\boldsymbol{\ell}_1, \dots, \boldsymbol{\ell}_T)$ from $\mathbb{R}^d$, independently of the learner's random

choices. For each round $t \in [T]$, the learner chooses an action $\boldsymbol{x}_t \in \mathcal{K}$ and subsequently observes the scalar $\lambda_t(\boldsymbol{x}_t) = \langle \boldsymbol{\ell}_t, \boldsymbol{x}_t \rangle$. The loss is assumed to be uniformly bounded in absolute value by a constant $L > 0$, *i.e.* $|\langle \boldsymbol{\ell}_t, \boldsymbol{x} \rangle| \leq L$ for all $t \in [T]$ and $\boldsymbol{x} \in \mathcal{K}$. Fixing the loss scale allows us to define meaningful metrics for evaluating the strategies employed by the learner, as detailed hereafter [13].

2.2. Regret Minimization

First, we formalize the notion of strategy mentioned in the preceding section. Let $H_t := (\boldsymbol{x}_k, y_k)_{k=1}^t$ denote the *random interaction history* generated up to round $t \leq T$, taking values in $\mathcal{H}_t := \big(\mathcal{K} \times [-L, L]\big)^t$. Moreover, let $\mathcal{F}_t := \sigma(H_t)$ be the smallest σ-algebra with respect to which H_t is measurable. Following Lattimore and Szepesvári [20], we define a *randomized strategy* as a sequence $(\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_T)$ of probability kernels $\boldsymbol{\xi}_t : \mathcal{H}_{t-1} \times \mathcal{B}(\mathcal{K}) \to [0, 1]$ such that $\boldsymbol{\xi}_t(\mathcal{H}_{t-1}, \cdot\,) \subseteq \Delta(\mathcal{K})$. We further require the $\mathcal{B}(\mathcal{H}_{t-1})$-measurability of $\boldsymbol{\xi}_t(\,\cdot\,, \mathcal{X})$ for every Borel set $\mathcal{X} \in \mathcal{B}(\mathcal{K})$, thereby ensuring that $\boldsymbol{\xi}_t(H_{t-1}, \mathcal{X})$ is in turn $\mathcal{F}_{t-1}$-measurable. Accordingly, at every round the learner plays $\boldsymbol{x}_t \sim q_t$, where $q_t := \boldsymbol{\xi}_t(H_{t-1}, \cdot\,)$ is referred to as the *sampling distribution*. Whenever q_t is absolutely continuous with respect to the Lebesgue measure, we write $q_t(d\boldsymbol{x}) = q_t(\boldsymbol{x})\, d\boldsymbol{x}$ with a slight abuse of notation.

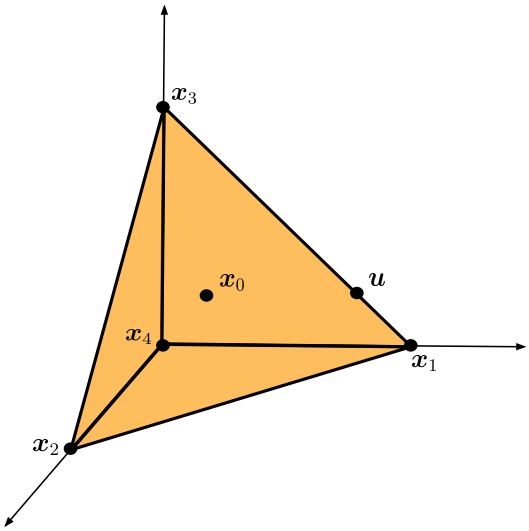


Figure 2.1: A simplex illustrates the distinction between multi-armed and linear bandits. In the multi-armed setting, the vertices correspond to the arms and are the only playable actions, while a point in their convex hull represents a distribution over them. The coefficients of the convex combination specify the sampling probabilities; the barycenter corresponds to the uniform distribution. In the linear setting, the entire simplex forms the action set: every convex combination of its vertices, including $\boldsymbol{u}$, is itself a playable action.

We evaluate a randomized strategy over rounds $t \in [T]$ through the notion of *regret*, which compares the learner's cumulative loss with that of the best fixed action *in hindsight*, *i.e.*

$$R_T(\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_T) := \sum_{t=1}^{T} \lambda_t(\boldsymbol{x}_t) - \min_{\boldsymbol{u} \in \mathcal{K}} \sum_{t=1}^{T} \lambda_t(\boldsymbol{u}) = \max_{\boldsymbol{u} \in \mathcal{K}} \sum_{t=1}^{T} \langle \boldsymbol{\ell}_t, \boldsymbol{x}_t - \boldsymbol{u} \rangle.$$

Randomization enables *regret minimization* under bandit feedback. Indeed, given any sampling distribution q_t and realized interaction $(\boldsymbol{x}_t, y_t)$, the learner can construct an estimate $\widehat{\boldsymbol{\ell}}_t = \widehat{\boldsymbol{\ell}}_t(\boldsymbol{x}_t, y_t, q_t)$ that yields $\widehat{\lambda}_t(\boldsymbol{x}) = \langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x} \rangle$ for all $\boldsymbol{x} \in \mathcal{K}$. This estimate may then be used to update the strategy in pursuit of this objective. At the same time, q_t must ensure sufficient exploration in order to control the estimation error $\langle \boldsymbol{\ell}_t - \widehat{\boldsymbol{\ell}}_t, \cdot \rangle$, giving rise to the well-known *exploration-exploitation trade-off* [2]. Since R_T is random, it is often convenient to decouple the analysis of the *estimated regret* $\widehat{R}_T$, obtained by replacing the true losses with their estimates, from the control of this estimation error.

Algorithm 2.1 Template Algorithm

Require $(\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_T)$.
Initialize $H_0 = \varnothing$.
for $t = 1, \dots, T$ **do**
  Set $q_t(\,\cdot\,) = \boldsymbol{\xi}_t(H_{t-1}, \cdot\,)$.
  Sample $\boldsymbol{x}_t \sim q_t(\,\cdot\,)$.
  Observe $y_t = \langle \boldsymbol{\ell}_t, \boldsymbol{x}_t \rangle$.
  Construct $\widehat{\boldsymbol{\ell}}_t$ from $(\boldsymbol{x}_t, y_t, q_t)$.
  Set $H_t = (H_{t-1}, (\boldsymbol{x}_t, y_t))$.
end for

2.3. Algorithmic Guarantees

Our goal in the analysis is to determine whether some strategy $(\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_T)$ admits a uniform upper bound on R_T over all admissible loss sequences, namely a *regret guarantee*. As previously mentioned, R_T is a random variable; hence these guarantees may be stated either in *expectation* or with *high probability*. For any horizon $T \in \mathbb{N}$, an expected regret guarantee takes the form $\mathbb{E}[R_T(\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_T)] \le B_T$, whereas a high-probability guarantee takes the form $\mathbb{P}(R_T(\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_T) \le B_{T,\delta}) \ge 1 - \delta$ for every confidence level $\delta \in (0, 1)$. The deterministic bounds B_T and $B_{T,\delta}$ may depend on the horizon and on fixed problem parameters, such as dimension and loss scale; $B_{T,\delta}$ may additionally depend on δ. A strategy is said to enjoy a *no-regret guarantee* if the corresponding bound is $o(T)$, *i.e.* if

the upper bound on the average regret vanishes as $T \to \infty$. For adversarial linear bandits, efficient strategies typically achieve a dependence of order $\widetilde{O}(\sqrt{T})$.

Regret over the entire horizon T may conceal poor performance on shorter intervals. For instance, let T be even and consider two actions $\boldsymbol{u}_1$, $\boldsymbol{u}_2$. Suppose that $\boldsymbol{u}_1$ incurs $\lambda_t(\boldsymbol{u}_1) = 1$ during the first $\frac{T}{2}$ rounds and $\lambda_t(\boldsymbol{u}_1) = 0$ thereafter, while the second action follows the opposite pattern. A learner that always selects the first action has zero regret over the entire horizon, since both actions incur a cumulative loss of $\frac{T}{2}$. However, over the first $\frac{T}{2}$ rounds, its regret against the second action is $\frac{T}{2}$. To account for these cases, we consider regret over nonempty intervals $I = [t_1, t_2] \subseteq [T]$ against a fixed comparator $\boldsymbol{u} \in \mathcal{K}$, namely

$$R_I(\boldsymbol{u}; \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_T) := \sum_{t \in I} \lambda_t(\boldsymbol{x}_t) - \sum_{t \in I} \lambda_t(\boldsymbol{u}) = \sum_{t \in I} \langle \boldsymbol{\ell}_t, \boldsymbol{x}_t - \boldsymbol{u} \rangle. \tag{2.1}$$

We call this quantity the *interval regret* and write $\mathcal{I}$ for the set of nonempty intervals in $[T]$. By compactness of $\mathcal{K}$, an optimal comparator exists and since the loss sequence is oblivious, it can be chosen deterministically on each fixed interval. We are interested in *weakly adaptive* strategies requiring, respectively in expectation and with high probability

$$\sup_{I \in \mathcal{I}} \mathbb{E}\left[\max_{\boldsymbol{u} \in \mathcal{K}} R_I(\boldsymbol{u}; \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_T)\right] \leq B_T, \tag{2.2}$$

$$\mathbb{P}\left(\sup_{I \in \mathcal{I}} \max_{\boldsymbol{u} \in \mathcal{K}} R_I(\boldsymbol{u}; \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_T) \leq B_{T,\delta}\right) \geq 1 - \delta, \tag{2.3}$$

where both B_T and $B_{T,\delta}$ are $o(T)$ for fixed problem parameters and (2.3) must hold for every $\delta \in (0, 1)$.

3 | Algorithms

In this chapter, we introduce an extension to exponential weights and the ingredients for weakly adaptive learning. We first describe the loss estimator and use minimum volume enclosing ellipsoids to build an exploration measure that controls its variance. We then introduce uniform fixed-share, which preserves probability mass across the decision set and enables comparison with any fixed action on an arbitrary interval. Finally, we add a pointwise correction that controls the comparator's estimation error with high probability.

3.1. Exponential Weights

Exponential weights is one of the fundamental decision-making principles in online learning. It consists in maintaining a distribution over the action set and concentrating it towards actions with smaller cumulative estimated losses through a multiplicative update [2, 12]. We focus on a continuous extension [6, 10] that employs probability densities: starting from $p_1(\boldsymbol{x}) = \frac{1}{\mathrm{vol}(\mathcal{K})}$, each round we set $p_{t+1}(\boldsymbol{x}) \propto p_t(\boldsymbol{x}) \exp(-\eta \widehat{\lambda}_t(\boldsymbol{x}))$ for all $\boldsymbol{x} \in \mathcal{K}$, up to a normalization constant. Here, $\eta > 0$ denotes the *learning rate* controlling the magnitude of each update. However, the learner does not exploit p_t directly and requires an additional exploration distribution ν. The sampling distribution q_t is therefore constructed through explicit exploration, *i.e.* by setting $q_t(d\boldsymbol{x}) := (1-\gamma)p_t(\boldsymbol{x})\, d\boldsymbol{x} + \gamma\nu(d\boldsymbol{x})$. The parameter $\gamma \in (0,1)$ is the *exploration rate* balancing the aforementioned exploration-exploitation trade-off by convexification. Let $\boldsymbol{M}_t := \mathbb{E}_{\boldsymbol{x}\sim q_t}[\boldsymbol{x} \otimes \boldsymbol{x}]$ denote the second-moment matrix of q_t. At each round, the learner constructs the estimator $\widehat{\boldsymbol{\ell}}_t := y_t \boldsymbol{M}_t^{-1} \boldsymbol{x}_t$, where the correction induced by $\boldsymbol{M}_t^{-1}$ accounts for the anisotropy[1] of the sampling distribution q_t, making the estimator conditionally unbiased given $\mathcal{F}_{t-1}$. It remains to choose ν so that $\boldsymbol{M}_t$ is nonsingular and sufficiently well-conditioned, ensuring exploration along all relevant directions while keeping the regret due to mixing under control.

[1] Here, *anisotropy* means that the sampling distribution has different second moments along different directions, thus exploring some directions more strongly than others.

3.2. Löwner-John Exploration

The distribution $\nu(\,\cdot\,)$ should be aware of the orientation and directional extent of the decision set, so that the γ-fraction of $q_t(\,\cdot\,)$ devoted to exploration is allocated intelligibly. For this sake, we employ the minimum volume enclosing ellipsoid of $\mathcal{K}$, also known as the *Löwner-John ellipsoid* [18], and denote it by $\mathcal{E}(\mathcal{K})$. Its unique parametrization, *i.e.* center $\boldsymbol{c} \in \mathbb{R}^d$ and shape matrix $\boldsymbol{H} \succ 0$, is obtained by solving the semi-infinite concave program

$$\min_{\boldsymbol{c}\in\mathbb{R}^d} \min_{\boldsymbol{H}\succ 0} \log\det(\boldsymbol{H}) \qquad \text{s.t.} \qquad \langle \boldsymbol{x}-\boldsymbol{c}, \boldsymbol{H}^{-1}(\boldsymbol{x}-\boldsymbol{c})\rangle \leq 1, \quad \forall\, \boldsymbol{x} \in \mathcal{K}. \tag{3.1}$$

As discussed in Section 1.1, computing such an ellipsoid can be demanding and we do not claim that the following exploration scheme can be constructed efficiently for general convex bodies. Furthermore, in order to preserve the original observation model, we employ a slight variation of Problem 3.1 that fixes $\boldsymbol{c} = 0$, effectively restricting the problem to origin-centered ellipsoids [20]. Given the optimizer $\boldsymbol{H}^\star$, the affine transformation $\mathcal{L} : \boldsymbol{x} \mapsto (\boldsymbol{H}^\star)^{-1/2}\boldsymbol{x}$ that maps $\mathcal{E}(\mathcal{K})$ onto $\mathbb{B}_2^d$ is well defined, and the image set $\mathcal{L}(\mathcal{K})$ is contained in $\mathbb{B}_2^d$. Such a configuration is referred to as *Löwner-John position*, and the following theorem guarantees the existence of a probability distribution supported on the contact points in $\mathcal{L}(\mathcal{K}) \cap \partial\mathbb{B}_2^d$.

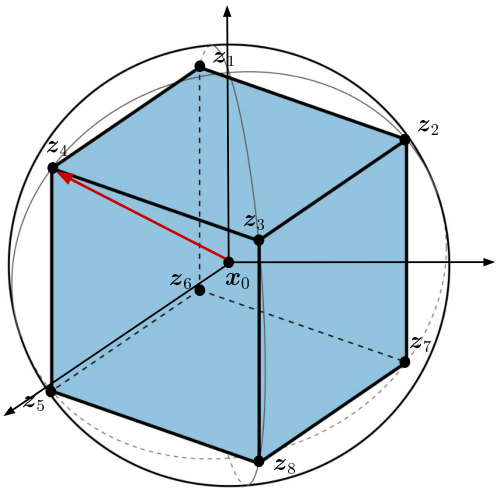


Figure 3.1: Löwner position for a cubic decision set in $\mathbb{R}^3$. The transformed set $\mathcal{L}(\mathcal{K})$ is inscribed in the Euclidean unit ball $\mathbb{B}_2^3$, which represents its minimum volume enclosing ellipsoid. The vertices $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_8$ are contact points with $\partial\mathbb{B}_2^3$ and support a probability distribution yielded by $\pi_1, \ldots, \pi_m$. The arrow in red illustrates an exploratory action toward a contact point, starting from $\boldsymbol{x}_0$ (the barycenter of $\mathbb{B}_2^3$ and the coordinate origin). This construction encourages exploration on geometrically informative boundary directions.

Theorem 3.1 (John's Theorem). *Let $\mathcal{K} \subset \mathbb{R}^d$ be a nonempty, full-dimensional, compact convex set and let $\mathcal{L}(\mathcal{K})$ denote its Löwner-John position. Then, there exist contact points $\mathcal{Z} := \{\boldsymbol{z}_1, \dots, \boldsymbol{z}_m\} \subseteq \mathcal{L}(\mathcal{K}) \cap \partial\mathbb{B}_2^d$, with $d \leq m \leq \frac{d(d+1)}{2}$ and strictly positive weights $\pi_1, \dots, \pi_m$ summing to one, such that $\sum_{i=1}^m \pi_i \boldsymbol{z}_i \otimes \boldsymbol{z}_i = \frac{\boldsymbol{I}_d}{d}$. The contact points support a discrete probability measure $\nu(\boldsymbol{z}_i) = \pi_i$ whose second-moment matrix is proportional to $\boldsymbol{I}_d$.*

Let ν be the probability distribution described in Theorem 3.1, with the second-moment matrix $\boldsymbol{G} := \mathbb{E}_{\boldsymbol{z}\sim\nu}[\boldsymbol{z} \otimes \boldsymbol{z}]$ satisfying $\boldsymbol{G} = \frac{\boldsymbol{I}_d}{d}$. Since $\mathcal{L}(\mathcal{K}) \subseteq \mathbb{B}_2^d$, we have $\|\boldsymbol{x}\|^2_{\boldsymbol{G}^{-1}} \leq d$ for every $\boldsymbol{x} \in \mathcal{L}(\mathcal{K})$, with equality at each contact point. Kiefer and Wolfowitz show that d is the smallest attainable supremum of this quadratic form over the decision set among probability measures with nonsingular second-moment matrices [19]. For this reason, ν is called $\boldsymbol{G}$*-optimal*. Therefore, we have obtained an exploration criterion that depends only on the geometry of the decision set and is independent of the loss sequence. Using ν in the mixture of Section 3.1 gives $\boldsymbol{M}_t \succeq \gamma\boldsymbol{G}$, ensuring that $\boldsymbol{M}_t$ is nonsingular and that $\|\boldsymbol{x}\|^2_{\boldsymbol{M}_t^{-1}} \leq \frac{d}{\gamma}$ for every $\boldsymbol{x} \in \mathcal{L}(\mathcal{K})$ and $t \in [T]$. Under the corresponding transformation of loss vectors into $(\boldsymbol{H}^\star)^{1/2}\boldsymbol{\ell}_t$, scalar losses and regret are fully preserved. Henceforth, we denote $\mathcal{L}(\mathcal{K})$ by $\mathcal{K}$, implicitly assuming the preprocessing described in this section.

3.3. Uniform Fixed-Share

The exponential weights density p_t depends on cumulative estimated losses, so actions penalized on earlier intervals may retain very little weight even when they subsequently become optimal. *Fixed-share* addresses this issue by redistributing a fixed fraction of probability mass after each multiplicative update [16]. We use the uniform-restart variant: let $\mu := \frac{1}{\mathrm{vol}(\mathcal{K})}$ denote the uniform density on $\mathcal{K}$ and let $\widetilde{p}_{t+1}$ be the density obtained from the multiplicative update at round t. For a share parameter $\alpha \in (0, 1)$, we set $p_{t+1} := (1-\alpha)\widetilde{p}_{t+1} + \alpha\mu$. Together with $p_1 = \mu$, this update guarantees the pointwise lower bound $p_t(\boldsymbol{x}) \geq \alpha\mu(\boldsymbol{x})$ for every $t \in [T]$ and $\boldsymbol{x} \in \mathcal{K}$, independently of the past loss sequence. Thus, every measurable region retains at least an α-fraction of its uniform probability mass. In Chapter 4, this lower bound controls the initial relative entropy in a *potential bound*, enabling comparison with a fixed action over intervals beginning at any round.

3.4. Pointwise Correction

The conditional unbiasedness of $\widehat{\boldsymbol{\ell}}_t$ makes the estimation errors introduced in Section 2.2 vanish in expectation. High-probability guarantees also require control of their cumulative fluctuations: overestimating the comparator's loss can cause the estimated regret to

understate the true regret. Following the third construction of Zimmert and Lattimore [26], we therefore introduce a downward bias calibrated to the uncertainty of each loss estimate. For every $\boldsymbol{x} \in \mathcal{K}$, we define the *pointwise correction* $b_t(\boldsymbol{x}) := \eta L^2 \|\boldsymbol{x}\|^2_{\boldsymbol{M}_t^{-1}}$ and the *corrected loss estimator* $\widehat{g}_t(\boldsymbol{x}) := \langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x} \rangle - b_t(\boldsymbol{x})$. Here, $L^2 \|\boldsymbol{x}\|^2_{\boldsymbol{M}_t^{-1}}$ bounds the conditional second moment of the estimated loss, so the correction acts as an optimism bonus scaled to estimation uncertainty. It applies to every action, including the comparator, lowering its estimated loss according to the estimation uncertainty induced by q_t. Chapter 4 formalizes this cancellation and controls the correction's average contribution under the learner's density.

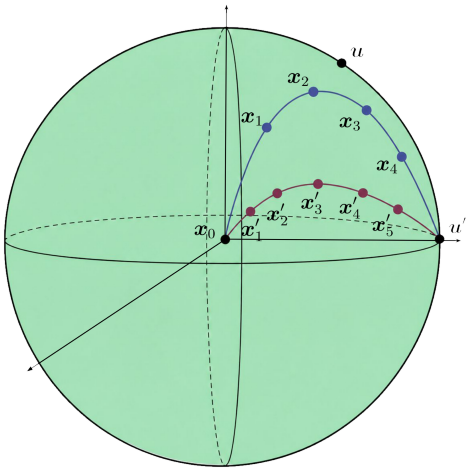


Figure 3.2: Intuitively, uniform fixed-share helps the learner revisit downweighted regions through uniform draws, whose mean is the barycenter of $\mathcal{K}$. From the common initialization $\boldsymbol{x}_0$, the schematic trajectories $\boldsymbol{x}'_1, \ldots, \boldsymbol{x}'_5$ (bordeaux, with fixed-share) and $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_4$ (blue, without fixed-share) initially move towards the comparator $\boldsymbol{u}$ and then turn towards $\boldsymbol{u}'$ after a loss change. The fixed-share trajectory shows a conservative response to the initial losses, reflecting the goal of attaining lower interval regret and comparable static regret.

3.5. Pseudocodes

Algorithm 3.1 follows the construction outlined by Luo et al. [22], combining exponential weights with explicit exploration and uniform fixed-share. The authors state an expected switching regret guarantee of $\widetilde{O}(d\sqrt{ST})$ without giving its proof. The next chapter establishes an expected interval regret guarantee and isolates the estimation term that obstructs a direct high-probability extension. Algorithm 3.2 addresses this term through the pointwise correction introduced above. To the best of our knowledge, no prior work

provides an algorithm with a $\widetilde{O}(\sqrt{T})$ weakly adaptive regret guarantee with high probability in the assumed setting.

Algorithm 3.1 Exponential Weights with Fixed Share

Input: $\mathcal{K}$, T, $L>0$, μ uniform density, ν Löwner-John measure, $\gamma\in(0,1)$, $\alpha\in(0,1)$, $\eta>0$.
Init: $p_1(\boldsymbol{x}) = \mu(\boldsymbol{x})$, $\forall \boldsymbol{x} \in \mathcal{K}$.
for $t = 1, \dots, T$ **do**
  Compute $q_t(d\boldsymbol{x}) = (1-\gamma)p_t(\boldsymbol{x})\,d\boldsymbol{x} + \gamma\nu(d\boldsymbol{x})$ and $\boldsymbol{M}_t = \mathbb{E}_{\boldsymbol{x}\sim q_t}[\boldsymbol{x}\otimes\boldsymbol{x}]$.
  Sample $\boldsymbol{x}_t \sim q_t$.
  Play $\boldsymbol{x}_t$ and observe $y_t = \langle \boldsymbol{\ell}_t, \boldsymbol{x}_t\rangle$.
  Compute $\widehat{\boldsymbol{\ell}}_t = y_t \boldsymbol{M}_t^{-1}\boldsymbol{x}_t$.
  Update

$$\widetilde{p}_{t+1}(\boldsymbol{x}) = \frac{p_t(\boldsymbol{x})\exp(-\eta\langle\widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle)}{\displaystyle\int_{\mathcal{K}} d\boldsymbol{z}\, p_t(\boldsymbol{z})\exp(-\eta\langle\widehat{\boldsymbol{\ell}}_t, \boldsymbol{z}\rangle)}.$$

  Set $p_{t+1}(\boldsymbol{x}) = (1-\alpha)\widetilde{p}_{t+1}(\boldsymbol{x}) + \alpha\mu(\boldsymbol{x})$.
end for

Algorithm 3.2 Corrected Exponential Weights with Fixed Share

Input: $\mathcal{K}$, T, $L>0$, μ uniform density, ν Löwner-John measure, $\gamma\in(0,1)$, $\alpha\in(0,1)$, $\eta>0$.
Init: $p_1(\boldsymbol{x}) = \mu(\boldsymbol{x})$, $\forall \boldsymbol{x} \in \mathcal{K}$.
for $t = 1, \dots, T$ **do**
  Compute $q_t(d\boldsymbol{x}) = (1-\gamma)p_t(\boldsymbol{x})\,d\boldsymbol{x} + \gamma\nu(d\boldsymbol{x})$ and $\boldsymbol{M}_t = \mathbb{E}_{\boldsymbol{x}\sim q_t}[\boldsymbol{x}\otimes\boldsymbol{x}]$.
  Sample $\boldsymbol{x}_t \sim q_t$.
  Play $\boldsymbol{x}_t$ and observe $y_t = \langle \boldsymbol{\ell}_t, \boldsymbol{x}_t\rangle$.
  Compute $\widehat{\boldsymbol{\ell}}_t = y_t \boldsymbol{M}_t^{-1}\boldsymbol{x}_t$ and $b_t(\boldsymbol{x}) = \eta L^2\|\boldsymbol{x}\|^2_{\boldsymbol{M}_t^{-1}}$.
  Set $\widehat{g}_t(\boldsymbol{x}) = \langle\widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle - b_t(\boldsymbol{x})$.
  Update

$$\widetilde{p}_{t+1}(\boldsymbol{x}) = \frac{p_t(\boldsymbol{x})\exp(-\eta\widehat{g}_t(\boldsymbol{x}))}{\displaystyle\int_{\mathcal{K}} d\boldsymbol{z}\, p_t(\boldsymbol{z})\exp(-\eta\widehat{g}_t(\boldsymbol{z}))}.$$

  Set $p_{t+1}(\boldsymbol{x}) = (1-\alpha)\widetilde{p}_{t+1}(\boldsymbol{x}) + \alpha\mu(\boldsymbol{x})$.
end for

4 | Analysis

In this chapter, we analyze the algorithms introduced in Chapter 3 and establish their weakly adaptive regret guarantees. We derive interval potential bounds through comparator smoothing and uniform fixed-share. We obtain a guarantee in expectation and examine the limitations of the same argument for high-probability guarantees. Finally, we combine the corrected potential with martingale concentration to control regret simultaneously over all intervals, making explicit the contributions of exploration and the pointwise correction.

4.1. Interval Regret Decompositions

A Dirac comparator has infinite relative entropy with respect to a continuous density, so it cannot enter the potential argument directly. Following Bubeck et al. [7], we replace it with the uniform density on a contracted copy of $\mathcal{K}$ centered at the comparator. The next two results bound the resulting relative entropy and the associated regret cost.

Lemma 4.1 (Comparator Smoothing). *For any fixed $\boldsymbol{u} \in \mathcal{K}$ and $\rho \in (0,1)$, define the homothety $h_{\boldsymbol{u},\rho} : \boldsymbol{x} \mapsto (1-\rho)\boldsymbol{u} + \rho\boldsymbol{x}$. Let $\Delta_{\boldsymbol{u},\rho}$ and μ be the uniform densities on $h_{\boldsymbol{u},\rho}(\mathcal{K})$ and $\mathcal{K}$, respectively. The relative entropy of $\Delta_{\boldsymbol{u},\rho}$ with respect to μ satisfies this identity*

$$D_{\mathrm{KL}}(\Delta_{\boldsymbol{u},\rho} \,\|\, \mu) := \int_{\mathcal{K}} d\boldsymbol{x}\, \Delta_{\boldsymbol{u},\rho}(\boldsymbol{x}) \log \frac{\Delta_{\boldsymbol{u},\rho}(\boldsymbol{x})}{\mu(\boldsymbol{x})} = d \log \frac{1}{\rho}. \tag{4.1}$$

Proof. Convexity gives $h_{\boldsymbol{u},\rho}(\mathcal{K}) \subseteq \mathcal{K}$, and the homothety scales its volume by a factor ρ^d. Since $\Delta_{\boldsymbol{u},\rho}(\boldsymbol{x}) = \frac{1}{\mathrm{vol}(h_{\boldsymbol{u},\rho}(\mathcal{K}))}$ on its support and $\mu(\boldsymbol{x}) = \frac{1}{\mathrm{vol}(\mathcal{K})}$ on $\mathcal{K}$, we have $\frac{\Delta_{\boldsymbol{u},\rho}(\boldsymbol{x})}{\mu(\boldsymbol{x})} = \rho^{-d}$ on the support of $\Delta_{\boldsymbol{u},\rho}$. Integrating the logarithm of this ratio against $\Delta_{\boldsymbol{u},\rho}$ gives (4.1). □

Proposition 4.1 (Comparator Smoothing Cost). *For any fixed $\boldsymbol{u} \in \mathcal{K}$ and $\rho \in (0,1)$, let $\Delta_{\boldsymbol{u},\rho}$ be the uniform density on $h_{\boldsymbol{u},\rho}(\mathcal{K})$ introduced in Lemma 4.1. For every round $t \in [T]$*

$$R_t(\boldsymbol{u}) \le R_t(\Delta_{\boldsymbol{u},\rho}) + 2\rho L. \tag{4.2}$$

Proof. Adding and subtracting the smoothed comparator loss gives

$$R_t(\boldsymbol{u}) = \langle \boldsymbol{\ell}_t, \boldsymbol{x}_t - \boldsymbol{u} \rangle = \mathbb{E}_{\boldsymbol{w}\sim\Delta_{\boldsymbol{u},\rho}} \langle \boldsymbol{\ell}_t, \boldsymbol{x}_t - \boldsymbol{w} \rangle + \mathbb{E}_{\boldsymbol{w}\sim\Delta_{\boldsymbol{u},\rho}} \langle \boldsymbol{\ell}_t, \boldsymbol{w} - \boldsymbol{u} \rangle$$
$$= R_t(\Delta_{\boldsymbol{u},\rho}) + \mathbb{E}_{\boldsymbol{w}\sim\Delta_{\boldsymbol{u},\rho}} \langle \boldsymbol{\ell}_t, \boldsymbol{w} - \boldsymbol{u} \rangle.$$

By Lemma 4.1, if $\boldsymbol{v}\sim\mu$ and $\boldsymbol{w}=h_{\boldsymbol{u},\rho}(\boldsymbol{v})$, then $\boldsymbol{w} \sim \Delta_{\boldsymbol{u},\rho}$ and $\boldsymbol{w}-\boldsymbol{u} = \rho(\boldsymbol{v}-\boldsymbol{u})$. Thus

$$\mathbb{E}_{\boldsymbol{w}\sim\Delta_{\boldsymbol{u},\rho}} \langle \boldsymbol{\ell}_t, \boldsymbol{w} - \boldsymbol{u} \rangle = \rho\, \mathbb{E}_{\boldsymbol{v}\sim\mu} \langle \boldsymbol{\ell}_t, \boldsymbol{v} - \boldsymbol{u} \rangle \le \rho\big(L - (-L)\big) = 2\rho L. \qquad \square$$

By Section 3.2, $\|\boldsymbol{x}\|_t^2 \le \frac{d}{\gamma}$ for every $\boldsymbol{x} \in \mathcal{K}$. The feedback bound from Section 2.1 and Cauchy-Schwarz give, for all $t \in [T]$ and $\boldsymbol{x} \in \mathcal{K}$

$$|\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x} \rangle| = |y_t|\, |\langle \boldsymbol{M}_t^{-1} \boldsymbol{x}_t, \boldsymbol{x} \rangle| = |y_t|\, |\langle \boldsymbol{M}_t^{-\frac{1}{2}} \boldsymbol{x}_t, \boldsymbol{M}_t^{-\frac{1}{2}} \boldsymbol{x} \rangle| \le L \|\boldsymbol{x}_t\|_t \|\boldsymbol{x}\|_t \le \frac{dL}{\gamma}. \tag{4.3}$$

We first derive an interval potential bound for Algorithm 3.1. We then apply the same argument to Algorithm 3.2, keeping the correction terms explicit for the high-probability analysis.

Lemma 4.2 (Potential Bound)**.** *For Algorithm 3.1, assume $\eta|\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x} \rangle| \le 1$ for all $t \in [T]$ and $\boldsymbol{x} \in \mathcal{K}$. For every interval $I = [t_1, t_2] \in \mathcal{I}$, comparator $\boldsymbol{u} \in \mathcal{K}$ and $\rho \in (0, 1)$, we have*

$$\widehat{R}_I(\Delta_{\boldsymbol{u},\rho}) - \widehat{R}_I(p_I) \le \frac{d}{\eta} \log \frac{1}{\rho} + \frac{1}{\eta} \log \frac{1}{\alpha}\, \mathbf{1}_{\{t_1 > 1\}} + \frac{|I|}{\eta} \log \frac{1}{1-\alpha} + \frac{\eta L^2}{1-\gamma} \sum_{t\in I} \|\boldsymbol{x}_t\|_t^2. \tag{4.4}$$

Proof. Let

$$\widehat{R}_t(\Delta_{\boldsymbol{u},\rho}) - \widehat{R}_t(p_t) := \langle \widehat{\boldsymbol{\ell}}_t, \mathbb{E}_{\boldsymbol{x}\sim p_t} \boldsymbol{x} - \mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}} \boldsymbol{x} \rangle. \tag{4.5}$$

We use Z_t to denote the normalization constant in the multiplicative update of Algorithm 3.1. For any density r with finite relative entropy with respect to p_t

$$D_{\mathrm{KL}}(r \,\|\, \widetilde{p}_{t+1}) = D_{\mathrm{KL}}(r \,\|\, p_t) + \eta \mathbb{E}_{\boldsymbol{x}\sim r} \langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x} \rangle + \log Z_t.$$

Using (1.1) with $\eta|\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x} \rangle| \le 1$ gives

$$\log Z_t \le \log \int_{\mathcal{K}} d\boldsymbol{x}\, p_t(\boldsymbol{x}) \left(1 - \eta \langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x} \rangle + \eta^2 \langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x} \rangle^2\right)$$
$$\le -\eta \mathbb{E}_{\boldsymbol{x}\sim p_t} \langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x} \rangle + \eta^2 \mathbb{E}_{\boldsymbol{x}\sim p_t} \langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x} \rangle^2.$$

Since $p_{t+1} \geq (1-\alpha)\widetilde{p}_{t+1}$, we have

$$D_{\mathrm{KL}}(r \,\|\, p_{t+1}) \leq D_{\mathrm{KL}}(r \,\|\, \widetilde{p}_{t+1}) + \log \frac{1}{1-\alpha}.$$

Rearranging the preceding inequalities gives

$$\begin{aligned} \mathbb{E}_{\boldsymbol{x}\sim p_t}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle - \mathbb{E}_{\boldsymbol{x}\sim r}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle \leq \frac{D_{\mathrm{KL}}(r \,\|\, p_t) - D_{\mathrm{KL}}(r \,\|\, p_{t+1})}{\eta} + \\ + \frac{1}{\eta}\log\frac{1}{1-\alpha} + \eta \mathbb{E}_{\boldsymbol{x}\sim p_t}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle^2. \end{aligned} \tag{4.6}$$

By explicit exploration, $\boldsymbol{M}_t \succeq (1-\gamma)\mathbb{E}_{\boldsymbol{x}\sim p_t}[\boldsymbol{x}\otimes\boldsymbol{x}]$. Therefore

$$\begin{aligned} \mathbb{E}_{\boldsymbol{x}\sim p_t}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle^2 = \langle \widehat{\boldsymbol{\ell}}_t, \mathbb{E}_{\boldsymbol{x}\sim p_t}[\boldsymbol{x}\otimes\boldsymbol{x}]\widehat{\boldsymbol{\ell}}_t\rangle \leq \frac{1}{1-\gamma}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{M}_t\widehat{\boldsymbol{\ell}}_t\rangle \\ = \frac{y_t^2}{1-\gamma}\langle \boldsymbol{M}_t^{-1}\boldsymbol{x}_t, \boldsymbol{M}_t\boldsymbol{M}_t^{-1}\boldsymbol{x}_t\rangle \leq \frac{L^2}{1-\gamma}\|\boldsymbol{x}_t\|_t^2. \end{aligned} \tag{4.7}$$

Set $r = \Delta_{\boldsymbol{u},\rho}$ in (4.6) and sum over I

$$\widehat{R}_I(\Delta_{\boldsymbol{u},\rho}) - \widehat{R}_I(p_I) \leq \frac{D_{\mathrm{KL}}(\Delta_{\boldsymbol{u},\rho} \,\|\, p_{t_1}) - D_{\mathrm{KL}}(\Delta_{\boldsymbol{u},\rho} \,\|\, p_{t_2+1})}{\eta} + \frac{|I|}{\eta}\log\frac{1}{1-\alpha} + \frac{\eta L^2}{1-\gamma}\sum_{t\in I}\|\boldsymbol{x}_t\|_t^2.$$

The second relative entropy is nonnegative. Since $p_1 = \mu$, fixed-share and Lemma 4.1 yield

$$D_{\mathrm{KL}}(\Delta_{\boldsymbol{u},\rho} \,\|\, p_{t_1}) \leq D_{\mathrm{KL}}(\Delta_{\boldsymbol{u},\rho} \,\|\, \mu) + \log\frac{1}{\alpha}\,\mathbf{1}_{\{t_1>1\}} = d\log\frac{1}{\rho} + \log\frac{1}{\alpha}\,\mathbf{1}_{\{t_1>1\}}. \qquad \square$$

Lemma 4.3 (Corrected Potential Bound)**.** *For Algorithm 3.2, assume $\gamma \geq 2\eta dL$ and $\eta \leq \frac{1}{4dL}$. For every interval $I = [t_1, t_2] \in \mathcal{I}$, comparator $\boldsymbol{u} \in \mathcal{K}$ and $\rho \in (0,1)$, we have*

$$\begin{aligned} \widehat{R}_I(\Delta_{\boldsymbol{u},\rho}) - \widehat{R}_I(p_I) \leq \frac{d}{\eta}\log\frac{1}{\rho} + \frac{1}{\eta}\log\frac{1}{\alpha}\,\mathbf{1}_{\{t_1>1\}} + \frac{|I|}{\eta}\log\frac{1}{1-\alpha} + \\ + \sum_{t\in I}\mathbb{E}_{\boldsymbol{x}\sim p_t} b_t(\boldsymbol{x}) - \sum_{t\in I}\mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}} b_t(\boldsymbol{x}) + \\ + \eta\sum_{t\in I}\mathbb{E}_{\boldsymbol{x}\sim p_t}\left(\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle - b_t(\boldsymbol{x})\right)^2. \end{aligned} \tag{4.8}$$

Proof. By (4.3) and the definition of b_t

$$\eta\left|\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle - b_t(\boldsymbol{x})\right| \leq \eta|\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle| + \eta b_t(\boldsymbol{x}) \leq \frac{\eta dL}{\gamma} + \frac{\eta^2 dL^2}{\gamma} \leq \frac{1}{2} + \frac{\eta L}{2} \leq \frac{1}{2} + \frac{1}{8d} < 1.$$

The derivation of (4.6) applies with $\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x} \rangle$ replaced by $\widehat{g}_t(\boldsymbol{x})$. Set $r = \Delta_{\boldsymbol{u},\rho}$, sum over I, and bound the second entropy as in Lemma 4.2. Expanding $\widehat{g}_t$ gives (4.8). □

Under the assumptions of Lemma 4.3, the exploration bound gives

$$2\eta b_t(\boldsymbol{x}) = 2\eta^2 L^2 \|\boldsymbol{x}\|_t^2 \le \frac{2\eta^2 d L^2}{\gamma} \le \eta L \le \frac{1}{4d} \le 1.$$

Thus $2\eta b_t(\boldsymbol{x})^2 \le b_t(\boldsymbol{x})$. Moreover

$$\mathbb{E}_{\boldsymbol{x}\sim p_t} b_t(\boldsymbol{x}) = \eta L^2 \operatorname{tr}\left(\boldsymbol{M}_t^{-1} \mathbb{E}_{\boldsymbol{x}\sim p_t}[\boldsymbol{x}\otimes\boldsymbol{x}]\right) \le \frac{\eta L^2}{1-\gamma} \operatorname{tr}(\boldsymbol{M}_t^{-1}\boldsymbol{M}_t) = \frac{\eta d L^2}{1-\gamma}. \tag{4.9}$$

Combining (1.1), the preceding bias bounds and (4.7) gives

$$\begin{aligned} \eta \sum_{t\in I} \mathbb{E}_{\boldsymbol{x}\sim p_t} \left(\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle - b_t(\boldsymbol{x})\right)^2 &\le \frac{2\eta L^2}{1-\gamma} \sum_{t\in I} \|\boldsymbol{x}_t\|_t^2 + \sum_{t\in I} \mathbb{E}_{\boldsymbol{x}\sim p_t} b_t(\boldsymbol{x}) \\ &\le \frac{2\eta L^2}{1-\gamma} \sum_{t\in I} \|\boldsymbol{x}_t\|_t^2 + \frac{\eta d L^2 |I|}{1-\gamma}. \end{aligned} \tag{4.10}$$

4.2. Guarantees in Expectation

Taking conditional expectations in Lemma 4.2 converts estimated loss differences into true ones by unbiasedness and replaces each leverage score with its conditional mean, $\mathbb{E}_{t-1}\|\boldsymbol{x}_t\|_t^2 = d$ at every round. We add the exploration and smoothing costs and choose parameters independent of the interval. For $T=1$, the deterministic regret bound is $2L$.

Theorem 4.1 (Expected Interval Regret)**.** *Let $T \ge 2$. With the parameters in* (4.14)*, Algorithm 3.1 guarantees, uniformly over all intervals $I \in \mathcal{I}$*

$$\mathbb{E}\Big[\max_{\boldsymbol{u}\in\mathcal{K}} R_I(\boldsymbol{u})\Big] \le O\Big(dL\sqrt{T\log T}\Big). \tag{4.11}$$

Proof. Fix $I = [t_1, t_2] \in \mathcal{I}$ and $\boldsymbol{u} \in \mathcal{K}$. Since p_t is $\mathcal{F}_{t-1}$-measurable, (1.4) and (4.5) give

$$\mathbb{E}_{t-1}\left[\widehat{R}_t(\Delta_{\boldsymbol{u},\rho}) - \widehat{R}_t(p_t)\right] = \langle \boldsymbol{\ell}_t, \mathbb{E}_{\boldsymbol{x}\sim p_t}\boldsymbol{x} - \mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}}\boldsymbol{x}\rangle.$$

Explicit exploration costs at most $2\gamma L$ per round. Taking expectations and applying Proposition 4.1 yields

$$\mathbb{E}[R_I(\boldsymbol{u})] \le \mathbb{E}\left[\widehat{R}_I(\Delta_{\boldsymbol{u},\rho}) - \widehat{R}_I(p_I)\right] + 2L|I|(\gamma+\rho). \tag{4.12}$$

By (1.2), $\mathbb{E}_{t-1}\|\boldsymbol{x}_t\|_t^2 = d$. Moreover, (4.3) gives $\eta|\langle\widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle| \le \frac{\eta dL}{\gamma} \le 1$ whenever $\gamma \ge \eta dL$. Applying Lemma 4.2 therefore yields

$$\mathbb{E}[R_I(\boldsymbol{u})] \le \frac{d}{\eta}\log\frac{1}{\rho} + \frac{1}{\eta}\log\frac{1}{\alpha}\mathbf{1}_{\{t_1>1\}} + \frac{|I|}{\eta}\log\frac{1}{1-\alpha} + \frac{\eta dL^2|I|}{1-\gamma} + 2L|I|(\gamma+\rho). \quad (4.13)$$

Let $\Lambda = d\log T + \log(T+1) + 1$ and choose

$$\rho = \frac{1}{T}, \qquad \alpha = \frac{1}{T+1}, \qquad \gamma = \eta dL, \qquad \eta = \min\left\{\frac{1}{2dL}, \sqrt{\frac{\Lambda}{dL^2T}}\right\}. \quad (4.14)$$

Then $\gamma \le \frac{1}{2}$ and

$$d\log\frac{1}{\rho} + \log\frac{1}{\alpha}\mathbf{1}_{\{t_1>1\}} + |I|\log\frac{1}{1-\alpha} \le d\log T + \log(T+1) + T\log\left(1+\frac{1}{T}\right) \le \Lambda,$$

$$\mathbb{E}[R_I(\boldsymbol{u})] \le \frac{\Lambda}{\eta} + 4\eta dL^2T + 2L.$$

If $\eta = \sqrt{\frac{\Lambda}{dL^2T}}$, this gives

$$\mathbb{E}[R_I(\boldsymbol{u})] \le 5L\sqrt{dT\Lambda} + 2L = O\Big(dL\sqrt{T\log T}\Big).$$

Otherwise, $T \le 4d\Lambda$, and $R_I(\boldsymbol{u}) \le 2L|I| \le 2LT \le 4L\sqrt{dT\Lambda}$ gives the same rate. Compactness and the oblivious loss sequence yield a deterministic optimal comparator $\boldsymbol{u}_I^\star$ for each I. Applying the bound to $\boldsymbol{u}_I^\star$ proves (4.11), since $R_I(\boldsymbol{u}_I^\star) = \max_{\boldsymbol{u}\in\mathcal{K}} R_I(\boldsymbol{u})$. □

The expected bound does not control the comparator estimation error on individual trajectories. For fixed $I \in \mathcal{I}$, $\boldsymbol{u} \in \mathcal{K}$, $\rho \in (0,1)$, and $\delta_2 \in (0,1)$, the Bernstein bound (4.17) below, Jensen's inequality, and (1.5) give, whenever $\gamma \ge \eta dL$

$$\begin{aligned}\sum_{t\in I}\langle\widehat{\boldsymbol{\ell}}_t - \boldsymbol{\ell}_t, \mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}}\boldsymbol{x}\rangle &\le \eta L^2\sum_{t\in I}\mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}}\|\boldsymbol{x}\|_t^2 + \frac{1}{\eta}\log\frac{1}{\delta_2}\\ &\le \frac{\eta dL^2|I|}{\gamma} + \frac{1}{\eta}\log\frac{1}{\delta_2},\end{aligned}$$

with probability at least $1-\delta_2$. For $\gamma = \eta dL$, the first term is $L|I|$ and has no counterpart in the uncorrected potential. The correction b_t cancels the preceding quadratic term. This limitation of the argument does not imply an impossibility result for the uncorrected algorithm.

4.3. Guarantees with High Probability

We use the history *filtration* $(\mathcal{F}_t)_{t=0}^T$ from Section 2.2. A process $(X_t)_{t=1}^T$ is *adapted* if X_t is $\mathcal{F}_t$-measurable; for an integrable adapted process, $X_t - \mathbb{E}_{t-1}X_t$ is a *martingale difference*. The bounds below control sampling, estimation, and stability; see Exercises 20.7 and 5.15 of Lattimore and Szepesvári [20]. We apply them on a fixed interval, using the pointwise correction to cancel the comparator estimation term. A union bound makes the guarantee uniform.

Proposition 4.2 (Azuma-Hoeffding Inequality)**.** *Let $(X_t)_{t=1}^T$ be adapted to $(\mathcal{F}_t)_{t=0}^T$ with $|X_t| \le L$ almost surely. For fixed $I \in \mathcal{I}$ and $\delta \in (0,1)$, with probability at least $1-\delta$*

$$\sum_{t\in I}(X_t - \mathbb{E}_{t-1}X_t) \le L\sqrt{2|I|\log\frac{1}{\delta}}. \tag{4.15}$$

Proposition 4.3 (Bernstein-Type Inequalities)**.** *Let $(X_t)_{t=1}^T$ be square-integrable and adapted to $(\mathcal{F}_t)_{t=0}^T$. Fix $I \in \mathcal{I}$, $\delta \in (0,1)$, and a deterministic $\eta > 0$. If $\eta(X_t - \mathbb{E}_{t-1}X_t) \le 1$ almost surely for all $t \in I$, then, with probability at least $1-\delta$*

$$\sum_{t\in I}(X_t - \mathbb{E}_{t-1}X_t) \le \eta\sum_{t\in I}\mathbb{E}_{t-1}(X_t - \mathbb{E}_{t-1}X_t)^2 + \frac{1}{\eta}\log\frac{1}{\delta}. \tag{4.16}$$

If instead $\eta X_t \le 1$ almost surely for all $t \in I$, then, with probability at least $1-\delta$

$$\sum_{t\in I}(X_t - \mathbb{E}_{t-1}X_t) \le \eta\sum_{t\in I}\mathbb{E}_{t-1}X_t^2 + \frac{1}{\eta}\log\frac{1}{\delta}. \tag{4.17}$$

Theorem 4.2 (High-Probability Interval Regret)**.** *Let $T \ge 2$ and $\delta \in (0,1)$. With the parameters in* (4.26) *and* (4.27)*, Algorithm 3.2 guarantees, with probability at least $1-\delta$, simultaneously for every $I \in \mathcal{I}$*

$$\max_{\boldsymbol{u}\in\mathcal{K}} R_I(\boldsymbol{u}) \le O\left(L\sqrt{dT\left(d\log T + \log\frac{T}{\delta}\right)}\right). \tag{4.18}$$

Proof. Fix a deterministic interval $I = [t_1, t_2] \in \mathcal{I}$, comparator $\boldsymbol{u} \in \mathcal{K}$, and $\rho \in (0,1)$.

Assume $\gamma \geq 2\eta dL$ and $\eta \leq \frac{1}{4dL}$. Adding and subtracting the losses at the means of q_t and p_t, then applying Proposition 4.1 and (4.5), gives

$$\begin{aligned} R_I(\boldsymbol{u}) \leq \sum_{t\in I}\langle \boldsymbol{\ell}_t, \boldsymbol{x}_t - \mathbb{E}_{\boldsymbol{x}\sim q_t}\boldsymbol{x}\rangle + \left(\widehat{R}_I(\Delta_{\boldsymbol{u},\rho}) - \widehat{R}_I(p_I)\right) + \\ + \sum_{t\in I}\langle \boldsymbol{\ell}_t - \widehat{\boldsymbol{\ell}}_t, \mathbb{E}_{\boldsymbol{x}\sim p_t}\boldsymbol{x} - \mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}}\boldsymbol{x}\rangle + 2L|I|(\gamma+\rho). \end{aligned} \tag{4.19}$$

Let $\delta_0, \delta_1, \delta_2, \delta_3 \in (0,1)$. Apply (4.15) to $X_t = \langle \boldsymbol{\ell}_t, \boldsymbol{x}_t\rangle$. With probability at least $1-\delta_0$

$$\sum_{t\in I}\langle \boldsymbol{\ell}_t, \boldsymbol{x}_t - \mathbb{E}_{\boldsymbol{x}\sim q_t}\boldsymbol{x}\rangle \leq L\sqrt{2|I|\log\frac{1}{\delta_0}}. \tag{4.20}$$

We bound the learner and comparator estimation errors separately, then control the stability term.

Learner estimation error. Use (4.17) with $X_t = -\mathbb{E}_{\boldsymbol{x}\sim p_t}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle$. By (4.3), $\eta X_t \leq \frac{\eta dL}{\gamma} \leq \frac{1}{2}$. Conditional unbiasedness, Jensen's inequality, and (4.7) give, with probability at least $1-\delta_1$

$$\begin{aligned} \sum_{t\in I}\langle \boldsymbol{\ell}_t - \widehat{\boldsymbol{\ell}}_t, \mathbb{E}_{\boldsymbol{x}\sim p_t}\boldsymbol{x}\rangle &\leq \eta\sum_{t\in I}\mathbb{E}_{t-1}\left(\mathbb{E}_{\boldsymbol{x}\sim p_t}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle\right)^2 + \frac{1}{\eta}\log\frac{1}{\delta_1} \\ &\leq \eta\sum_{t\in I}\mathbb{E}_{t-1}\mathbb{E}_{\boldsymbol{x}\sim p_t}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle^2 + \frac{1}{\eta}\log\frac{1}{\delta_1} \\ &\leq \frac{\eta L^2}{1-\gamma}\sum_{t\in I}\mathbb{E}_{t-1}\|\boldsymbol{x}_t\|_t^2 + \frac{1}{\eta}\log\frac{1}{\delta_1} = \frac{\eta dL^2|I|}{1-\gamma} + \frac{1}{\eta}\log\frac{1}{\delta_1}, \end{aligned} \tag{4.21}$$

where the last equality follows from (1.2).

Comparator estimation error. Take $X_t = \mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle$. Again, (4.3) gives $\eta X_t \leq \frac{1}{2}$. By (4.17), Jensen's inequality, and (1.5), with probability at least $1-\delta_2$

$$\begin{aligned} \sum_{t\in I}\langle \widehat{\boldsymbol{\ell}}_t - \boldsymbol{\ell}_t, \mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}}\boldsymbol{x}\rangle &\leq \eta\sum_{t\in I}\mathbb{E}_{t-1}\left(\mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle\right)^2 + \frac{1}{\eta}\log\frac{1}{\delta_2} \\ &\leq \eta\sum_{t\in I}\mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}}\mathbb{E}_{t-1}\langle \widehat{\boldsymbol{\ell}}_t, \boldsymbol{x}\rangle^2 + \frac{1}{\eta}\log\frac{1}{\delta_2} \\ &\leq \eta L^2\sum_{t\in I}\mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}}\|\boldsymbol{x}\|_t^2 + \frac{1}{\eta}\log\frac{1}{\delta_2} \\ &= \sum_{t\in I}\mathbb{E}_{\boldsymbol{x}\sim\Delta_{\boldsymbol{u},\rho}}b_t(\boldsymbol{x}) + \frac{1}{\eta}\log\frac{1}{\delta_2}. \end{aligned} \tag{4.22}$$

Stability term. The exploration bound from Section 3.2 gives $\eta L\|\boldsymbol{x}_t\|_t^2 \le \frac{\eta d L}{\gamma} \le 1$. Moreover

$$\eta^2 L^3 \|\boldsymbol{x}_t\|_t^4 = \Big(\eta L\|\boldsymbol{x}_t\|_t^2\Big)\Big(\eta L^2\|\boldsymbol{x}_t\|_t^2\Big) \le \eta L^2\|\boldsymbol{x}_t\|_t^2.$$

Apply (4.17) to $X_t = L\|\boldsymbol{x}_t\|_t^2$ using the same parameter η, which is admissible by the preceding bound. Multiplying the inequality by $\frac{\eta L}{1-\gamma}$ gives, with probability at least $1-\delta_3$

$$\begin{aligned}\frac{\eta L^2}{1-\gamma}\sum_{t\in I}\Big(\|\boldsymbol{x}_t\|_t^2 - \mathbb{E}_{t-1}\|\boldsymbol{x}_t\|_t^2\Big) &\le \frac{\eta^2 L^3}{1-\gamma}\sum_{t\in I}\mathbb{E}_{t-1}\|\boldsymbol{x}_t\|_t^4 + \frac{L}{1-\gamma}\log\frac{1}{\delta_3}\\ &\le \frac{\eta L^2}{1-\gamma}\sum_{t\in I}\mathbb{E}_{t-1}\|\boldsymbol{x}_t\|_t^2 + \frac{L}{1-\gamma}\log\frac{1}{\delta_3}.\end{aligned}$$

Adding the conditional mean and using (1.2) yields

$$\begin{aligned}\frac{\eta L^2}{1-\gamma}\sum_{t\in I}\|\boldsymbol{x}_t\|_t^2 &\le \frac{2\eta L^2}{1-\gamma}\sum_{t\in I}\mathbb{E}_{t-1}\|\boldsymbol{x}_t\|_t^2 + \frac{L}{1-\gamma}\log\frac{1}{\delta_3}\\ &= \frac{2\eta d L^2|I|}{1-\gamma} + \frac{L}{1-\gamma}\log\frac{1}{\delta_3}.\end{aligned} \tag{4.23}$$

Consequently, (4.10) gives

$$\begin{aligned}\eta\sum_{t\in I}\mathbb{E}_{\boldsymbol{x}\sim p_t}\Big(\langle\widehat{\boldsymbol{\ell}}_t,\boldsymbol{x}\rangle - b_t(\boldsymbol{x})\Big)^2 &\le \frac{2\eta L^2}{1-\gamma}\sum_{t\in I}\|\boldsymbol{x}_t\|_t^2 + \frac{\eta d L^2|I|}{1-\gamma}\\ &\le \frac{5\eta d L^2|I|}{1-\gamma} + \frac{2L}{1-\gamma}\log\frac{1}{\delta_3}.\end{aligned} \tag{4.24}$$

Combining the bounds. On the intersection of these four events, the comparator correction in (4.22) cancels its negative counterpart in (4.8). Substituting (4.8), (4.9), (4.20), (4.21) and (4.24) into (4.19) gives, with probability at least $1-\delta_0-\delta_1-\delta_2-\delta_3$

$$\begin{aligned}R_I(\boldsymbol{u}) \le L\sqrt{2|I|\log\frac{1}{\delta_0}} + 2L|I|(\gamma+\rho) + \frac{d}{\eta}\log\frac{1}{\rho} + \frac{1}{\eta}\log\frac{1}{\alpha}\,\mathbf{1}_{\{t_1>1\}} +\\ + \frac{|I|}{\eta}\log\frac{1}{1-\alpha} + \frac{1}{\eta}\log\frac{1}{\delta_1} + \frac{1}{\eta}\log\frac{1}{\delta_2} + \frac{2L}{1-\gamma}\log\frac{1}{\delta_3} + \frac{7\eta d L^2|I|}{1-\gamma}.\end{aligned} \tag{4.25}$$

Parameter choice. Set

$$\Lambda_\delta = d\log T + \log(T+1) + 1 + \log\frac{1}{\delta_1} + \log\frac{1}{\delta_2} + \log\frac{1}{\delta_3}$$

and choose

$$\rho = \frac{1}{T}, \qquad \alpha = \frac{1}{T+1}, \qquad \gamma = 2\eta d L, \qquad \eta = \min\left\{\frac{1}{4dL}, \sqrt{\frac{\Lambda_\delta}{18dL^2T}}\right\}. \tag{4.26}$$

Then $\gamma \le \frac{1}{2}$, $\frac{2L}{1-\gamma} \le 4L \le \frac{1}{\eta}$, while $|I| \log \frac{1}{1-\alpha} \le 1$ and $2\rho L|I| \le 2L$. Thus (4.25) gives

$$R_I(\boldsymbol{u}) \le L\sqrt{2|I| \log \frac{1}{\delta_0}} + \frac{\Lambda_\delta}{\eta} + 18\eta d L^2 T + 2L.$$

If $\eta = \sqrt{\frac{\Lambda_\delta}{18dL^2T}}$

$$R_I(\boldsymbol{u}) \le L\sqrt{2|I| \log \frac{1}{\delta_0}} + 2L\sqrt{18dT\Lambda_\delta} + 2L.$$

Otherwise $T \le \frac{8}{9} d\Lambda_\delta$, and

$$R_I(\boldsymbol{u}) \le 2LT \le \frac{4\sqrt{2}}{3} L\sqrt{dT\Lambda_\delta}.$$

In either case

$$R_I(\boldsymbol{u}) \le O\left(L\sqrt{|I| \log \frac{1}{\delta_0}} + L\sqrt{dT\Lambda_\delta} + L\right).$$

Uniformity over intervals. Finally, choose

$$N_T := \frac{T(T+1)}{2}, \qquad \delta_0 = \delta_1 = \delta_2 = \delta_3 = \frac{\delta}{4N_T}. \tag{4.27}$$

For each interval I, select a deterministic minimizer $\boldsymbol{u}_I^\star \in \arg\min_{\boldsymbol{u}\in\mathcal{K}} \sum_{t\in I} \langle \boldsymbol{\ell}_t, \boldsymbol{u}\rangle$. As in Chapter 2, compactness and the oblivious loss sequence justify this choice. Apply the four concentration bounds to each pair $(I, \boldsymbol{u}_I^\star)$. A union bound over the N_T intervals gives total failure probability at most $4N_T \frac{\delta}{4N_T} = \delta$. On the resulting event, $R_I(\boldsymbol{u}) \le R_I(\boldsymbol{u}_I^\star)$ for every $\boldsymbol{u} \in \mathcal{K}$ and $I \in \mathcal{I}$. No union bound over the uncountable comparator set is required. Since $|I| \le T$ and

$$\Lambda_\delta = d\log T + \log(T+1) + 1 + 3\log\frac{4N_T}{\delta} = O\left(d\log T + \log\frac{T}{\delta}\right),$$

substituting into the regret bound proves (4.18) simultaneously over all intervals. □

5 | Conclusions

This thesis establishes that a single exponential weights learner can be compared with the best fixed action on any interval of an oblivious loss sequence, also with a guarantee that holds simultaneously over all intervals with high probability. The result brings together two requirements that arise at different points of the analysis: retaining enough probability mass when an interval begins and controlling estimation error at its comparator. It is proved for exact sampling and second moments, leaving an approximate implementation with explicit error bounds as a computational question. Extending the guarantee to adaptive adversaries is another question, relevant to using the algorithm as a primal learner in constrained online learning, where dual updates depend on past observations.

5.1. Acknowledgments

I would like to thank my supervisors for the exceptional opportunity to work with them over the past year and for their guidance throughout this time. This thesis grew out of such experience. I would also like to thank the Department of Electronics, Information and Bioengineering at Politecnico di Milano for providing a welcoming place to work.

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